Abhai Sai Nikhil Pothuraju

Data Engineer | Azure | Databricks | PySpark | Snowflake | SQL
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Professional Summary

Data Engineer with 3 years of data engineering experience across AT&T and IBM, building cloud ETL/ELT pipelines, lakehouse tables, and data warehouses with Python, SQL, Azure Databricks, PySpark, Delta Lake, Snowflake, dbt, and Apache Airflow. Delivered measurable outcomes in performance (41% faster daily processing), cost ($9K/month in verified AWS savings), and data reliability (99% on-time delivery before reporting cutoffs). Builds AI data infrastructure, including RAG ingestion and retrieval pipelines on Azure AI Search and Azure OpenAI that raised top-five retrieval hit rate from 69% to 86%. Experienced in incremental loading, data quality validation, infrastructure as code (Terraform), and CI/CD with GitHub Actions.

Technical Skills

Programming & APIs: Python, SQL, FastAPI, REST APIs
Data Platforms: Snowflake, Azure Databricks, Apache Spark (PySpark), Delta Lake, dbt, PostgreSQL
Streaming & CDC: Apache Kafka, Debezium, Spark Structured Streaming
Cloud – Azure: Azure Data Factory, Azure Databricks, Azure AI Search, Azure OpenAI
Cloud – AWS: Amazon S3, AWS Glue
Data Engineering: ETL/ELT Pipelines, Data Modeling, Star Schema, Data Warehousing, Lakehouse, Incremental Loading, Delta MERGE, Late-Arriving Data, Deduplication, Partitioned Parquet, Data Quality Validation, Source Reconciliation, Freshness Checks, Data Lineage, Restart-Safe Processing, Performance Optimization
Snowflake: External Stages, COPY INTO, MERGE, SQL Transformations, Effective-Dated Models
Orchestration & DevOps: Apache Airflow, Azure Data Factory, Terraform (Infrastructure as Code), Docker, GitHub Actions (CI/CD), Automated Testing, dbt Tests, Prometheus, Grafana
AI & ML Data: Retrieval-Augmented Generation (RAG), Hybrid Search, pgvector, Document Chunking, Vector Indexing, Metadata Enrichment, Retrieval Evaluation, LLM Tool Use and Evaluation, Point-in-Time Feature Engineering, Customer Segmentation (RFM, K-Means), Time-Series Forecasting (Holt-Winters)
BI & Analytics: Power BI, Streamlit, Dashboarding, KPI Scorecards, Cloud Cost Analytics
Security & Governance: Role-Based Access Control (RBAC), Version Tracking, Citation Validation, Audit Logging

Professional Experience

AT&T Chicago, IL
AI Data Engineer *May 2026 – Present*
Network Support Knowledge Retrieval (RAG)

- Built Python and Azure Data Factory ingestion pipelines for **8,500** approved troubleshooting guides and incident summaries, powering an Azure OpenAI assistant that cut median reference lookup time from **7 to 4 minutes** in a 35-person pilot.
- Raised top-five retrieval hit rate from **69% to 86%** on 180 SME-reviewed questions by refining document chunking, adding equipment-model metadata, and tuning retrieval in Azure AI Search.
- Implemented document fingerprints, version tracking, and deletion propagation for incremental indexing, making approved updates searchable within **2 hours** while processing **82% fewer documents** than full daily reindexing.
- Reduced responses citing superseded procedures from **11% to 3%** by adding active-version filters and citation checks.
- Integrated automated retrieval evaluations into the release workflow, gating index and retrieval changes against the SME-reviewed question set before deployment.
- Provisioned Azure AI Search and supporting storage with Terraform, maintaining version-controlled infrastructure-as-code configurations for repeatable deployments across development and test environments.

AT&T Remote
Data Engineer Intern *May 2025 – May 2026*
Repeat Service Visit Analytics

- Developed PySpark transformations in Azure Databricks to consolidate approximately **600,000** monthly appointment, work-order, and diagnostic records into Delta Lake service-history tables for daily field-operations reporting.
- Improved visit-to-repair-ticket linkage from **89% to 97%** by implementing deduplication, completion-code standardization, and Delta MERGE logic for late-arriving updates.
- Raised on-time completion before the 7 a.m. reporting cutoff from **92% to 99%** across six workflows by extending Azure Data Factory pipelines with freshness checks, source reconciliation, and restart-safe processing.

- Partnered with a data scientist to build **24** point-in-time features that excluded information unavailable at prediction time, preventing data leakage and cutting dataset preparation from **2 days to 3 hours** per training iteration.
- Prepared daily risk-score reporting tables and a Power BI scorecard for an eight-week scheduling pilot covering approximately **4,000** appointments, supporting reviews of appointment risk and repeat-visit outcomes.

IBM Hyderabad, India
Data Engineer *Jan 2023 – Aug 2024*
Internal Infrastructure Cost and Usage Analytics

- Built Python and AWS Glue pipelines to standardize billing and inventory data in Amazon S3 and load curated datasets into Snowflake via external stages and COPY INTO, replacing manual preparation for daily cost reporting.
- Developed dbt and Snowflake SQL models for billing and effective-dated cost-center ownership, increasing spend attributable to accountable owners from **81% to 96%** across approximately **$180,000** in monthly AWS usage.
- Joined billing and utilization data to identify idle development instances and unattached storage, supporting owner-approved cleanup that reduced spend by approximately **$9,000 per month**, verified across three consecutive billing cycles.
- Cut daily processing time from **105 to 62 minutes (41%)** using incremental extraction, partitioned Parquet, and Snowflake MERGE operations that incorporated new charges and billing corrections without duplicating records.
- Orchestrated AWS Glue ingestion, Snowflake loading, dbt transformations, and validation in Apache Airflow DAGs.
- Added dbt tests and SQL reconciliation checks for billing totals, duplicate charges, and ownership mappings, reducing monthly dataset sign-off from **3 business days to 1**.

Projects

DataOps Investigator – AI Data Reliability Platform | *Python, Kafka, Spark, dbt, Airflow, PostgreSQL, Claude API* GitHub

- Built a synthetic e-commerce data platform with batch and CDC ingestion (Debezium, Kafka), a Spark and Delta Lake lakehouse, dbt dimensional models with SCD2 history, and Airflow orchestration.
- Developed metadata collectors, dbt-manifest lineage, and a rule-based incident engine that correlates failed tests, skipped models, and reconciliation mismatches into one incident with downstream dashboard impact.
- Built a read-only LLM investigator with **11** allowlisted, parameterized tools and hybrid pgvector and full-text runbook retrieval; a validator rejects uncited or ungrounded claims and forces abstention when evidence is insufficient.
- Evaluated on **35** failure cases with Claude Haiku: correct root cause in **29–31** cases across three runs, **100%** valid citations in the final run, and zero forbidden actions.
- Added human-approved recovery actions with dry runs, audit logs, and post-recovery verification, served through FastAPI, Streamlit, and Prometheus/Grafana with GitHub Actions CI.

People Analytics AI Copilot | *Python, FastAPI, RBAC, Docker, GitHub Actions* GitHub

- Built an AI-powered HR analytics application in Python and FastAPI that lets users query workforce metrics through natural-language questions.
- Implemented controlled analytics workflows using predefined data-access tools and role-based access control (RBAC), backed by automated tests, Docker containerization, and GitHub Actions CI/CD.

RetailPulse – Retail Sales & Customer Analytics Platform | *Python, SQL, Power BI* GitHub | Dashboard

- Built an end-to-end retail analytics pipeline to clean, validate, transform, and model e-commerce data using Python and SQL.
- Implemented RFM and K-Means customer segmentation and Holt-Winters demand forecasting in Python.
- Designed a Power BI star-schema model and SQL query layer for sales, customer, and product analysis.

Education

University of Illinois Springfield Springfield, IL
Master of Science in Computer Science *Aug 2024 – May 2026*